

W1 – 4.3 Co-function Identities

MHF4U

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1) Simplify.

a) $\sin x \left(\frac{1}{\cos x} \right)$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

b) $(\cos x)(\sec x)$

$$= \cancel{\cos x} \left(\frac{1}{\cancel{\cos x}} \right)$$

$$= 1$$

c) $1 - \cos^2 x$

$$= \sin^2 x$$

d) $1 - \sin^2 x$

$$= \cos^2 x$$

e) $\frac{\tan x}{\sin x}$

$$= \frac{\left(\frac{\sin x}{\cos x} \right)}{\left(\frac{\sin x}{1} \right)}$$

$$= \left(\frac{\cancel{\sin x}}{\cos x} \right) \left(\frac{1}{\cancel{\sin x}} \right)$$

$$= \frac{1}{\cos x}$$

i) $\frac{\sin x \cos x}{1 - \sin^2 x}$

$$= \frac{\sin x \cancel{\cos x}}{\cos^2 x}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

f) $(1 - \sin x)(1 + \sin x)$

$$= 1 - \sin^2 x$$

$$= \cos^2 x$$

j) $\frac{1 - \cos^2 x}{\sin x \cos x}$

$$= \frac{\sin^2 x}{\cancel{\sin x} \cos x}$$

$$= \frac{1}{\cos x}$$

$$= \sec x$$

g) $\left(\frac{1}{\tan x} \right) \sin x$

$$= \frac{1}{\left(\frac{\sin x}{\cos x} \right)} (\sin x)$$

$$= \frac{\cos x}{\cancel{\sin x}} (\cancel{\sin x})$$

$$= \cos x$$

h) $\frac{1 + \tan^2 x}{\tan^2 x}$

$$= \frac{\frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x}}{\left(\frac{\sin^2 x}{\cos^2 x} \right)}$$

$$= \frac{\left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right)}{\left(\frac{\sin^2 x}{\cos^2 x} \right)}$$

$$= \frac{1}{\cancel{\cos^2 x}} \times \frac{\cancel{\cos^2 x}}{\sin^2 x}$$

$$= \frac{1}{\sin^2 x}$$

$$= \csc^2 x$$

2) Prove the following identities.

a) $\sin^2 x (1 + \cot^2 x) = 1$

LS

$$= \sin^2 x (1 + \cot^2 x)$$

GI

$$= \sin^2 x \left(1 + \frac{\cos^2 x}{\sin^2 x} \right)$$

$$= \sin^2 x + \frac{\cancel{\sin^2 x} \cos^2 x}{\cancel{\sin^2 x}}$$

$$= \sin^2 x + \cos^2 x$$

PI

$$= 1$$

LS = RS

RS

$$= 1$$

b) $1 - \cos^2 x = \tan x \cos x \sin x$

LS

$$= 1 - \cos^2 x$$

$$= \sin^2 x$$

RS

$$= \tan x \cos x \sin x$$

GI

$$= \left(\frac{\sin x}{\cancel{\cos x}} \right) (\cancel{\cos x}) (\sin x)$$

$$= \sin^2 x$$

LS = RS

c) $\cos x \tan^3 x = \sin x \tan^2 x$

$$\begin{array}{ll} \text{LS} & \text{RS} \\ = \cos x \tan^3 x & = \sin x \tan^2 x \\ = \cancel{\cos x} \left(\frac{\sin^3 x}{\cancel{\cos^3 x}} \right) & = \sin x \left(\frac{\sin^2 x}{\cos^2 x} \right) \\ = \frac{\sin^3 x}{\cos^2 x} & = \frac{\sin^3 x}{\cos^2 x} \end{array}$$

LS = RS

d) $1 - 2 \cos^2 \theta = \sin^4 \theta - \cos^4 \theta$

DOS: $a^2 - b^2$
 $= (a-b)(a+b)$

$$\begin{array}{ll} \text{LS} & \text{RS} \\ = 1 - 2 \cos^2 \theta & = (\sin^2 \theta)^2 - (\cos^2 \theta)^2 \\ & = (\sin^2 \theta - \cos^2 \theta)(\sin^2 \theta + \cos^2 \theta) \\ & = (\cancel{\sin^2 \theta} - \cos^2 \theta)(1) \\ & = \cancel{1} - \cos^2 \theta - \cos^2 \theta \\ & = 1 - 2 \cos^2 \theta \end{array}$$

LS = RS

e) $\cot x + \frac{\sin x}{1 + \cos x} = \csc x$

$$\begin{array}{ll} \text{LS} & \text{RS} \\ = \frac{\cancel{\cos x}}{(1 + \cancel{\cos x}) \sin x} + \frac{\sin x}{1 + \cos x} \cdot \frac{\sin x}{\sin x} & = \csc x \\ = \frac{\cancel{\cos x} (1 + \cos x) + \sin^2 x}{(1 + \cos x)(\sin x)} & = \frac{1}{\sin x} \\ = \frac{\cancel{\cos x} + \cos^2 x + \sin^2 x}{(1 + \cos x)(\sin x)} \text{PI} & \\ = \frac{\cancel{\cos x} + 1}{(1 + \cos x)(\sin x)} & \\ = \frac{1}{\sin x} & \end{array}$$

LS = RS

f) $\frac{\sec x}{\sin x} + \frac{\csc x}{\cos x} = \frac{2}{\sin x \cos x}$

$$\begin{array}{ll} \text{LS} & \text{RS} \\ = \frac{\sec x}{\sin x} + \frac{\csc x}{\cos x} & = \frac{2}{\sin x \cos x} \\ = \left(\frac{1}{\cos x} \right) + \left(\frac{1}{\sin x} \right) & \\ = \left(\frac{1}{\cos x} \right) \left(\frac{\sin x}{1} \right) + \left(\frac{1}{\sin x} \right) \left(\frac{\cos x}{1} \right) & \\ = \frac{1}{\sin x \cos x} + \frac{1}{\sin x \cos x} & \\ = \frac{2}{\sin x \cos x} & \end{array}$$

LS = RS

g) $\frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin x \cos x} = 1 - \tan x$

$$\begin{array}{ll} \text{LS} & \text{RS} \\ = \frac{(\cos x - \sin x)(\cancel{\cos x} + \sin x)}{\cos x (\cancel{\cos x} + \sin x)} & = 1 - \tan x \\ = \frac{\cos x - \sin x}{\cos x} & = \frac{\cos x}{\cos x} - \frac{\sin x}{\cos x} \\ & = \frac{\cos x - \sin x}{\cos x} \end{array}$$

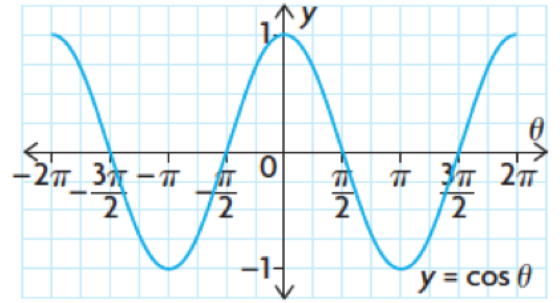
LS = RS

h) $\frac{1}{1 + \cos x} + \frac{1}{1 - \cos x} = 2 \csc^2 x$

$$\begin{array}{ll} \text{LS} & \text{RS} \\ = \frac{1}{1 + \cos x} + \frac{1}{1 - \cos x} & = 2 \csc^2 x \\ = \frac{1}{1 + \cos x} + \frac{1}{1 - \cos x} & = 2 \left(\frac{1}{\sin^2 x} \right) \\ = \frac{1 - \cancel{\cos x} + 1 + \cancel{\cos x}}{(1 + \cos x)(1 - \cos x)} & = \frac{2}{\sin^2 x} \\ = \frac{2}{1 - \cos^2 x} & \\ = \frac{2}{\sin^2 x} & \end{array}$$

LS = RS

3)a) Use transformations and the cosine function to write three equivalent expressions for the following graph:



① $y = \cos \theta$

② $y = \cos(\theta - 2\pi)$

③ $y = \cos(\theta + 2\pi)$

b) Transform your 3 equations from part a) to write the equation of 3 sine functions that represent the graph.

① $y = \sin(\theta + \frac{\pi}{2})$

② $y = \sin(\theta - 2\pi + \frac{\pi}{2}) = \sin(\theta - \frac{3\pi}{2})$

③ $y = \sin(\theta + 2\pi + \frac{\pi}{2}) = \sin(\theta + \frac{5\pi}{2})$

4) Use the co-function identities to write an expression that is equivalent to each of the following expressions.

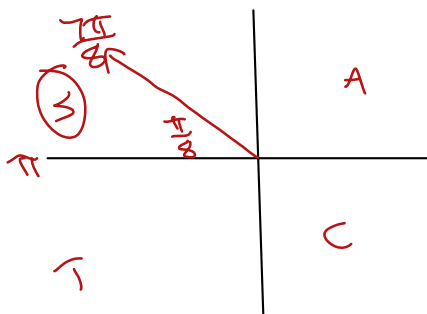
a) $\sin \frac{\pi}{6} = \cos(\frac{\pi}{2} - \frac{\pi}{6})$
 $= \cos(\frac{3\pi}{6} - \frac{\pi}{6})$
 $= \cos(\frac{2\pi}{6})$
 $= \cos(\frac{\pi}{3})$

b) $\cos \frac{5\pi}{12} = \sin(\frac{\pi}{2} - \frac{5\pi}{12})$
 $= \sin(\frac{6\pi}{12} - \frac{5\pi}{12})$
 $= \sin(\frac{\pi}{12})$

c) $\cos \frac{5\pi}{16} = \sin(\frac{\pi}{2} - \frac{5\pi}{16})$
 $= \sin(\frac{8\pi}{16} - \frac{5\pi}{16})$
 $= \sin(\frac{3\pi}{16})$

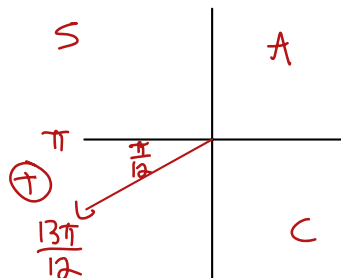
5) Write an expression that is equivalent to each of the following expressions, using the related acute angle.

a) $\sin \frac{7\pi}{8}$



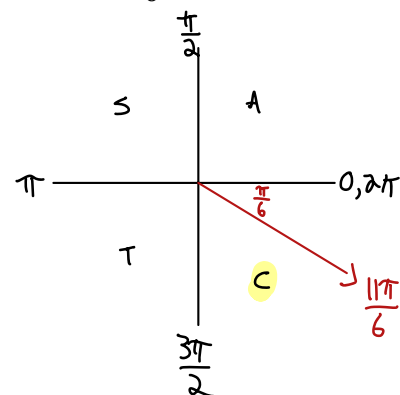
$\sin \frac{7\pi}{8} = \sin \frac{\pi}{8}$

b) $\cos \frac{13\pi}{12}$



$\cos \frac{13\pi}{12} = -\cos \frac{\pi}{12}$

c) $\cos \frac{11\pi}{6}$



$\cos \frac{11\pi}{6} = \cos \frac{\pi}{6}$

6) Given that $\sin \frac{\pi}{6} = \frac{1}{2}$, use an equivalent trigonometric expression to show that $\cos \frac{\pi}{3} = \frac{1}{2}$

$$\begin{aligned}\cos \frac{\pi}{3} &= \sin \left(\frac{\pi}{2} - \frac{\pi}{3} \right) \\ &= \sin \left(\frac{\pi}{6} \right) \\ &= \frac{1}{2}\end{aligned}$$

7) Given that $\sin \frac{\pi}{6} = \frac{1}{2}$, use an equivalent trigonometric expression to show that $\cos \frac{2\pi}{3} = -\frac{1}{2}$

$$\begin{aligned}\cos \frac{2\pi}{3} &= \sin \left(\frac{\pi}{2} - \frac{2\pi}{3} \right) \\ &= \sin \left(\frac{3\pi}{6} - \frac{4\pi}{6} \right) \\ &= \sin \left(-\frac{\pi}{6} \right)\end{aligned} \quad \begin{aligned}&\rightarrow = -\sin \frac{\pi}{6} \\ &= -\frac{1}{2}\end{aligned}$$

8) Given that $\csc \frac{\pi}{4} = \sqrt{2}$, use an equivalent trigonometric expression to show that $\sec \frac{3\pi}{4} = -\sqrt{2}$

$$\begin{aligned}\sec \frac{3\pi}{4} &= \frac{1}{\cos \frac{3\pi}{4}} \\ &= \frac{1}{\sin \left(\frac{\pi}{2} - \frac{3\pi}{4} \right)}\end{aligned} \quad \begin{aligned}&\rightarrow = \frac{1}{\sin \left(-\frac{\pi}{4} \right)} \\ &= \frac{1}{-\sin \frac{\pi}{4}} \\ &= -\csc \frac{\pi}{4} \\ &= -\sqrt{2}\end{aligned}$$

9) Given that $\cos \frac{3\pi}{11} \approx 0.6549$, use equivalent trigonometric expressions to evaluate the following, to four decimal places.

$$\begin{aligned}\text{a) } \sin \frac{5\pi}{22} &= \cos \left(\frac{\pi}{2} - \frac{5\pi}{22} \right) \\ &= \cos \left(\frac{11\pi}{22} - \frac{5\pi}{22} \right) \\ &= \cos \left(\frac{6\pi}{22} \right) \\ &= \cos \left(\frac{3\pi}{11} \right) \\ &\approx 0.6549\end{aligned}$$

$$\begin{aligned}\text{b) } \sin \frac{17\pi}{22} &= \cos \left(\frac{\pi}{2} - \frac{17\pi}{22} \right) \\ &= \cos \left(\frac{11\pi}{22} - \frac{17\pi}{22} \right) \\ &= \cos \left(\frac{-6\pi}{22} \right) \\ &= \cos \left(\frac{-3\pi}{11} \right) \\ &= \cos \left(\frac{3\pi}{11} \right) \\ &\approx 0.6549\end{aligned}$$

Answer Key

1)a) $\tan x$ **b)** 1 **c)** $\sin^2 x$ **d)** $\cos^2 x$ **e)** $\sec x$ **f)** $\cos^2 x$ **g)** $\cos x$ **h)** $\csc^2 x$ **i)** $\tan x$ **j)** $\tan x$

3) Answers will vary depending but possible solutions are:

a) $y = \cos(\theta + 2\pi), y = \cos(\theta - 2\pi), y = \cos(\theta - 4\pi)$

b) $y = \sin(\theta + \frac{5\pi}{2}), y = \sin(\theta - \frac{3\pi}{2}), y = \sin(\theta - \frac{7\pi}{2})$

4)a) $\cos \frac{\pi}{3}$ **b)** $\sin \frac{\pi}{12}$ **c)** $\sin \frac{3\pi}{16}$ **5)a)** $\sin \frac{\pi}{8}$ **b)** $-\cos \frac{\pi}{12}$ **c)** $\cos \frac{\pi}{6}$

9)a) $\sin \frac{5\pi}{22} = \cos \frac{3\pi}{11} \sim 0.6549$ **b)** $\sin \frac{17\pi}{22} = \cos \left(-\frac{3\pi}{11}\right) = \cos \left(\frac{3\pi}{11}\right) \sim 0.6549$